1 Where neutral hydrogen survives after reionisation

After reionisation completes at $z \approx 6$, the intergalactic medium is overwhelmingly ionised (neutral fraction $\lesssim 10^{-4}$). Surviving neutral hydrogen resides almost exclusively in *self-shielded, overdense gas clouds* — damped Lyman-α systems (DLAs) with column densities $N_{\rm HI} \geq 2 \times 10^{20}$ cm^{-2} — within the interstellar medium of galaxies hosted by dark matter haloes (Wolfe et al., 2005).

1.1 The critical halo-mass window

Pinetti et al. state that haloes of mass 5×10^9–10^{12} $M_\odot$ contain most of the cosmic HI. This range emerges from two physical boundary effects:

Low-mass cutoff ($M \lesssim 10^8$ $M_\odot$). Haloes below $\sim 10^8$ $M_\odot$ have virial (circular) velocities $v_{\rm c} \lesssim 25$–30 km s^{-1}. The metagalactic UV photoionisation background heats the IGM to $T \sim 10^4$ K, raising the Jeans mass and preventing gas from accreting into these shallow potential wells (Rees, 1986; Efstathiou, 1992; Pontzen et al., 2008). Simulations confirm that haloes with $v_{\rm c} < 30$ km s^{-1} do not efficiently host neutral gas (Bagla et al., 2010).

High-mass suppression ($M \gtrsim 10^{12}$ $M_\odot$). Haloes above $\sim 10^{12}$ $M_\odot$ have virial temperatures exceeding 10^5–10^6 K, where infalling gas is shock-heated to the hot, X-ray-emitting phase rather than cooling to form a neutral component. AGN feedback in these massive systems further heats and expels gas.

The intervening "sweet spot" at 10^{10}–10^{12} $M_\odot$ contains haloes deep enough to accrete gas despite the UV background, yet cool enough for that gas to settle into a neutral phase. ALFALFA abundance-matching analyses and the conditional HI mass function both confirm this picture (Papastergis et al., 2013; Padmanabhan et al., 2017).

1.2 The 21-cm hyperfine transition

The line arises from the spin-flip of the electron in the hydrogen $1S$ ground state between the triplet ($F = 1$, parallel spins, statistical weight 3) and singlet ($F = 0$, antiparallel, weight 1) hyperfine levels, separated by

$$\Delta E = 5.88 \times 10^{-6} \text{ eV}. \tag{1}$$

The rest-frame frequency is

$$\nu_{21} = 1420.405 \text{ MHz} \qquad (\lambda = 21.106 \text{ cm}), \tag{2}$$

and the spontaneous emission coefficient is $A_{10} = 2.85 \times 10^{-15}$ s^{-1}, corresponding to a natural lifetime of $\sim 10^7$ yr.

In the post-reionisation regime, the spin temperature T_s greatly exceeds $T_{\rm CMB}$ via collisional coupling and the Wouthuysen–Field (Lyα pumping) effect, so the 21-cm signal is always seen in *emission* against the CMB. The brightness temperature then becomes independent of T_s. The observed frequency maps directly to redshift via

$$\nu_{\rm obs} = \frac{1420.405}{1+z} \text{ MHz}, \tag{3}$$

which is the foundation of HI intensity mapping as a three-dimensional probe of large-scale structure (Bharadwaj & Ali, 2004, Pinetti et al. Eq. 3.1).

1.3 The HI density parameter $\Omega_{\rm HI}(z)$

Defined as

$$\Omega_{\rm HI}(z) = \frac{\rho_{\rm HI}(z)}{\rho_{\rm crit,0}}, \tag{4}$$

this parameter sets the overall amplitude of the HI signal. In the halo model it is computed as (Pinetti et al. Eq. 3.2)

$$\Omega_{\rm HI} = \frac{1}{\rho_{\rm crit}} \int \frac{{\rm d}n}{{\rm d}M} M_{\rm HI}(M,z) \, {\rm d}M. \tag{5}$$

Observational constraints come from multiple probes. Local 21-cm emission surveys such as HIPASS give $\Omega_{\rm HI} h_{70} \approx 3.8 \times 10^{-4}$ at $z \sim 0$ (Zwaan et al., 2005), while the ALFALFA $\alpha.100$ catalogue gives

$\Omega_{\rm HI}h_{70} \approx (3.9\text{–}4.3) \times 10^{-4}$ (Jones et al., 2018). DLA absorbers from the SDSS-III sample yield $\Omega_{\rm HI} \approx (0.8\text{–}1.0) \times 10^{-3}$ at $z \sim 2\text{–}3.5$ (Noterdaeme et al., 2012), and the GGG survey extends to $z \sim 4.9$ with $\Omega_{\rm HI} = (0.98 \pm 0.20) \times 10^{-3}$ (Crighton et al., 2015). Intensity mapping cross-correlations such as GBT × WiggleZ constrain $\Omega_{\rm HI}b_{\rm HI}$ at $z \sim 0.8$ (Switzer et al., 2013; Masui et al., 2013).

A commonly used empirical fit is $\Omega_{\rm HI}(z) \propto (1+z)^{0.6}$, indicating a gradual increase from $\sim 4 \times 10^{-4}$ at $z = 0$ to $\sim 10^{-3}$ at $z \sim 2\text{–}5$. The cross-correlation signal amplitude $C_\ell^{{\rm HI}\times\gamma}$ scales *linearly* with $\Omega_{\rm HI}$ through $\bar{T}_b$, so the $\sim 20\text{–}30\%$ observational uncertainty in $\Omega_{\rm HI}$ propagates directly into a comparable fractional uncertainty in the predicted signal.

1.4 Mean brightness temperature $\bar{T}_b(z)$

In the post-reionisation limit ($T_s \gg T_{\rm CMB}$), the mean 21-cm brightness temperature is (Pinetti et al. Eq. 3.4)

$$\boxed{\bar{T}_b(z) \approx 188\, h\, \Omega_{\rm HI}(z)\, \frac{(1+z)^2}{E(z)}\ {\rm mK},} \tag{6}$$

where $E(z) = H(z)/H_0$. Typical values are $\sim 55\ \mu$K at $z = 0$, $\sim 140\ \mu$K at $z = 1$, and $\sim 240\ \mu$K at $z = 2$, increasing with redshift because both $\Omega_{\rm HI}$ and $(1+z)^2/E(z)$ grow. These sub-millikelvin amplitudes — four to five orders of magnitude below the astrophysical foregrounds — are the fundamental challenge of intensity mapping.

2 The HI–halo mass relation and its parameter space

Pinetti et al. adopt the parametric relation of Padmanabhan et al. (2017) (their Eq. 3.7):

$$\boxed{M_{\rm HI}(M,z) = \alpha\, f_{H,c}\, M \left(\frac{M}{10^{11}\, h^{-1}\, M_\odot}\right)^{\beta} \exp\left[-\left(\frac{v_{\rm c,0}}{v_{\rm c}(M,z)}\right)^3\right],} \tag{7}$$

where $f_{H,c} = (1 - Y_p) \approx 0.76$ is the cosmic hydrogen mass fraction and $v_{\rm c}(M,z)$ is the halo circular velocity.

2.1 Physical meaning of each parameter

Normalisation α. Controls the overall fraction of baryonic hydrogen in neutral form per halo. It directly scales $\Omega_{\rm HI}$ and hence $\bar{T}_b$.

Power-law slope β. Negative ($\beta \approx -0.58$ to -0.69), encoding the decline of HI efficiency with halo mass: more massive haloes convert a smaller fraction of their baryons to neutral gas, reflecting virial shock-heating and AGN feedback.

Minimum circular velocity $v_{\rm c,0}$. Sets the low-mass cutoff. Haloes with $v_{\rm c} < v_{\rm c,0}$ have their HI content exponentially suppressed, encoding photoionisation heating that prevents gas accretion in shallow potential wells.

2.2 Parameter calibration

Padmanabhan et al. (2017) fit these parameters via MCMC (using CosmoHammer) simultaneously to: (i) the ALFALFA HI mass function at $z \sim 0$ (Martin et al., 2010); (ii) DLA column-density distributions at $z \sim 2\text{–}5$ (Noterdaeme et al., 2012; Crighton et al., 2015); (iii) $\Omega_{\rm HI}$ measurements across $z \sim 0\text{–}5$; (iv) DLA incidence rate ${\rm d}N/{\rm d}X$ and DLA bias $b_{\rm DLA}$; (v) small-scale clustering from ALFALFA (Martin et al., 2012; Papastergis et al., 2013); and (vi) intensity mapping constraints at $z \sim 0.8$ (Switzer et al., 2013).

The specific values adopted by Pinetti et al. are

$$\alpha = 0.176, \qquad \beta = -0.69, \qquad v_{\rm c,0} = 101.61\ {\rm km\,s^{-1}}. \tag{8}$$

These differ notably from the Padmanabhan et al. (2017) published best-fits for the altered-NFW profile ($\alpha = 0.17$, $\beta = -0.55$, $v_{\rm c,0} = 37.2\ {\rm km\,s^{-1}}$) or the exponential profile ($\alpha = 0.09$, $\beta = -0.58$, $v_{\rm c,0} =$

36.3 km s^{-1}). The notably larger $v_{c,0} = 101.61$ km s^{-1} implies a more aggressive low-mass cutoff, placing the HI minimum-mass halo at roughly $10^{10}\ M_\odot$ rather than $\sim 10^8$–$10^9\ M_\odot$. This choice shifts the effective $b_{\rm HI}$ upward while reducing the total $\Omega_{\rm HI}$ contribution from low-mass haloes, and warrants careful verification against the original parameter tables.

2.3 Alternative models

The principal competitors are cosmological hydrodynamical simulations. **IllustrisTNG** (Villaescusa-Navarro et al., 2018) assigns HI in post-processing using the Rahmati et al. (2013) self-shielding prescription; it agrees well with $\Omega_{\rm HI}$ at $z \sim 0$ but shows more complex $M_{\rm HI}(M)$ with significant scatter and environmental dependence (assembly bias) not captured by the parametric model. **SIMBA** (Davé et al., 2019, 2020) uses meshless-finite-mass hydrodynamics with torque-limited BH accretion and bipolar kinetic AGN jets, and tends to produce lower HI masses in intermediate-mass haloes than IllustrisTNG due to different AGN feedback prescriptions. **MUFASA** (Davé et al., 2017) is the predecessor of SIMBA with simpler feedback.

All models agree reasonably at $z \sim 0$ but diverge by factors of 2–5 at $z > 1$, particularly in $\Omega_{\rm HI}$ contributions from haloes above $10^{12}\ M_\odot$. The CHIME auto-power spectrum interpretation (CHIME Collaboration, 2026) found 3.1–4.0σ disagreement with IllustrisTNG predictions at $z \sim 1$, likely due to inadequate modelling of nonlinear redshift-space clustering of HI in the simulations.

2.4 Sensitivity analysis

On large scales, $C_\ell^{\rm HI\times\gamma} \propto \bar{T}_b \times b_{\rm HI} \times W_\gamma \times P_m$. Since $\bar{T}_b \propto \alpha$ linearly, α *is the parameter that most directly scales the overall signal amplitude*. The slope β redistributes HI across halo masses, affecting both $\Omega_{\rm HI}$ and the effective $b_{\rm HI}$ in a correlated but partially cancelling manner. The cutoff $v_{c,0}$ has the most complex effect: increasing it reduces $\Omega_{\rm HI}$ but raises $b_{\rm HI}$ (by removing low-bias, low-mass haloes), so the net impact on the product $\Omega_{\rm HI} \times b_{\rm HI}$ depends on the steepness of the halo mass function.

Fisher-forecast analyses (Padmanabhan et al., 2019) show the HI auto-power spectrum is most sensitive to $v_{c,0}$ and β as astrophysical parameters, while the cross-correlation amplitude uncertainty is dominated by the combined product $\Omega_{\rm HI} \times b_{\rm HI}$.

3 The HI density profile within haloes

Pinetti et al. model the HI radial density within haloes using the altered NFW profile with a thermal core (their Eq. 3.9):

$$\boxed{\rho_{\rm HI}(r) = \frac{\rho_0\, r_s^3}{(r + 0.75\, r_s)\,(r + r_s)^2},} \tag{9}$$

where $r_s = R_{\rm vir}/c_{\rm HI}$ is the scale radius. This profile was introduced by Maller & Bullock (2004) for multiphase halo gas and subsequently adopted for DLA studies by Barnes & Haehnelt (2010, 2014) and incorporated into the halo model by Padmanabhan et al. (2017).

3.1 Why a core, not a cusp

Unlike collisionless dark matter, which forms the familiar r^{-1} NFW cusp, neutral hydrogen gas has *thermal pressure support*. The warm neutral medium at $T \sim 10^4$ K exerts gas pressure that resists gravitational concentration toward the halo centre, producing a flat core at $r \lesssim 0.75\, r_s$. Additionally, gas retains angular momentum during infall, settling into disc-like configurations rather than cuspy spheroids. The profile transitions smoothly to the standard NFW r^{-3} envelope at large radii ($r \gg r_s$). The $0.75\, r_s$ core scale arises naturally from thermal equilibrium considerations in multiphase cooling simulations (Cole et al., 2000; Maller & Bullock, 2004).

3.2 HI concentration $c_{\rm HI}$

Padmanabhan et al. (2017) parametrise the HI concentration as

$$c_{\rm HI}(M, z) = c_{\rm HI,0} \left(\frac{M}{10^{11}\, M_\odot}\right)^{-0.109} \times \frac{4}{(1+z)^{0.13}}, \tag{10}$$

with a best-fit $c_{\mathrm{HI},0} \approx 139$ and redshift exponent $\gamma \approx 0.13$. This is roughly *ten times larger* than typical dark matter concentrations ($c_{\mathrm{DM}} \sim 10$–20; Macciò et al. 2007). The physical reason is straightforward: baryonic gas can *dissipate energy* through radiative cooling (atomic line emission, Compton cooling), losing kinetic energy and contracting to much smaller radii than collisionless dark matter. For a Milky-Way-like galaxy, the HI disc extends to $\sim$15–30 kpc while the virial radius is $\sim$200–350 kpc — a ratio directly mirroring the high c_{HI}.

3.3 Normalisation via mass conservation

The central density ρ_0 is uniquely determined by requiring (Pinetti et al. Eq. 3.10):

$$\int_0^{R_{\mathrm{vir}}} \rho_{\mathrm{HI}}(r)\, 4\pi r^2\, \mathrm{d}r = M_{\mathrm{HI}}(M, z). \tag{11}$$

This is simply mass conservation.

3.4 Fourier transform and angular-scale dependence

The normalised Fourier transform that enters the power spectrum calculation is (Pinetti et al. Eq. 3.14)

$$\tilde{u}_{\mathrm{HI}}(k \mid M, z) = \frac{1}{M_{\mathrm{HI}}} \int_0^{R_{\mathrm{vir}}} \rho_{\mathrm{HI}}(r)\, \frac{\sin(kr)}{kr}\, 4\pi r^2\, \mathrm{d}r. \tag{12}$$

Because $c_{\mathrm{HI}} \gg c_{\mathrm{DM}}$, the HI profile is far more compact and $\tilde{u}_{\mathrm{HI}} \approx 1$ out to much higher k than the DM NFW profile. On **large scales** ($\ell \lesssim 100$), the 2-halo term dominates and both profiles give $\tilde{u} \to 1$, so the profile shape is irrelevant. The profile choice becomes significant for the **1-halo term at** $\ell \gtrsim 200$–**500**, corresponding to $k \gtrsim 0.1$–$0.3\, h^{-1}$ Mpc at $z \sim 0.3$–1.

For the Pinetti et al. cross-correlation forecast, which spans $\ell \sim 10$–1000, profile details matter primarily at the high-ℓ end.

3.5 Robustness from simulations and observations

IllustrisTNG shows that mean HI profiles are approximately universal across mass and redshift but exhibit large halo-to-halo scatter (Villaescusa-Navarro et al., 2018). Resolved HI observations from the THINGS survey (Walter et al., 2008) and WHISP show that azimuthally averaged profiles flatten or decline near the centre and fall exponentially in the outer regions — better described by an *exponential disc* than a modified NFW at low redshift, as acknowledged by Padmanabhan et al. (2017). For the cross-correlation analysis at $z \sim 0.1$–0.5, the modified NFW remains a reasonable approximation since the profile primarily affects the 1-halo term at high ℓ where instrumental noise is large.

4 HI bias: how neutral hydrogen traces large-scale structure

The effective HI bias in the halo model is (Pinetti et al. Eq. 3.6):

$$\boxed{b_{\mathrm{HI}}(z) = \frac{1}{\bar{\rho}_{\mathrm{HI}}} \int \mathrm{d}M\, \frac{\mathrm{d}n}{\mathrm{d}M}\, M_{\mathrm{HI}}(M, z)\, b(M, z).} \tag{13}$$

This is a *mass-weighted* average of the halo bias, where the weighting is by HI content rather than number or total mass.

4.1 Predicted and measured values

The HI bias *increases monotonically with redshift*. At $z \sim 0$, HI traces relatively underdense regions (blue, late-type galaxies), giving $b_{\mathrm{HI}} < 1$. By $z \sim 2$–3, HI-hosting haloes become rarer peaks in the density field with higher bias, reaching $b_{\mathrm{HI}} \sim 2$–3.

Table 1: HI bias measurements and predictions at different redshifts.

Redshift	$b_{\rm HI}$ (predicted)	Observational constraint
$z \sim 0$	0.65–0.85	ALFALFA: $b_{\rm HI} \approx 0.8$ (Martin et al., 2012)
$z \sim 0.4$	0.8–1.0	MeerKAT $\times$ WiggleZ: $\Omega_{\rm HI} b_{\rm HI} r = (0.86 \pm 0.16) \times 10^{-3}$ (Cunnington et al., 2023)
$z \sim 0.8$	0.9–1.1	GBT $\times$ WiggleZ: $\Omega_{\rm HI} b_{\rm HI} r = (0.43 \pm 0.07) \times 10^{-3}$ (Masui et al., 2013)
$z \sim 2.3$	1.5–2.2	DLA bias: $b_{\rm DLA} = 2.17 \pm 0.2$ (Font-Ribera et al., 2012)

4.2 Scale dependence

Linear (scale-independent) bias breaks down at $k \sim 0.3\ h^{-1}\,$Mpc at $z \leq 3$ (Villaescusa-Navarro et al., 2018; Castorina & Villaescusa-Navarro, 2017). Using the Limber approximation ($\ell \sim k\chi(z)$) for $z \sim 0.3$–1 where $\chi \sim 1$–3 Gpc, this corresponds to $\ell \sim 300$–900. For the Pinetti et al. multipole range of $\ell = 10$–1000, the regime $\ell \lesssim 100$–300 is safely in the linear bias regime, while $\ell > 300$–500 enters the nonlinear regime where the 1-halo term and scale-dependent bias become important.

4.3 Impact on the cross-correlation

On large scales, $C_\ell^{\rm HI\times\gamma} \propto b_{\rm HI}$ linearly, so a 20% uncertainty in $b_{\rm HI}$ translates directly to a $\sim 20\%$ uncertainty in the astrophysical cross-correlation amplitude. Current constraints on $b_{\rm HI}$ have 20–50% uncertainties depending on redshift, making it one of the dominant astrophysical systematics alongside $M_{\rm HI}(M, z)$ and $\Omega_{\rm HI}$. However, joint measurement of HI auto- and cross-power spectra can break the $\Omega_{\rm HI}$–$b_{\rm HI}$ degeneracy, since

$$P_{\rm auto} \propto b_{\rm HI}^2, \qquad P_{\rm cross} \propto b_{\rm HI}. \tag{14}$$

5 Observational validation: from ALFALFA through MeerKAT and CHIME

5.1 The ALFALFA survey

The Arecibo Legacy Fast ALFA survey used the 7-beam ALFA receiver on the 305-m Arecibo telescope to perform a blind extragalactic HI survey over ~ 7000 deg^2 at $z < 0.06$. The final α.100 catalogue contains $\sim 31{,}500$ extragalactic HI sources (Haynes et al., 2018), spanning HI masses from $\sim 10^6$ to $\sim 10^{10.8}\ M_\odot$. Martin et al. (2012) and Papastergis et al. (2013) measured the correlation function and power spectrum of HI-selected galaxies, finding that HI sources cluster more weakly than optically selected samples, consistent with $b_{\rm HI} \approx 0.8$ at $z \sim 0$.

5.2 Pinetti et al. Figure 1 comparison

The halo-model prediction agrees well on **large scales** ($k < 1\ h^{-1}\,$Mpc), where the 2-halo term dominates, but systematically **underestimates clustering on small scales** (below $\sim 1\ h^{-1}\,$Mpc). This discrepancy arises because the standard halo model with a single continuous HI profile per halo does not capture satellite galaxy contributions, gas stripping in group environments, or other baryonic physics that redistribute HI on sub-halo scales. Padmanabhan et al. (2023) showed that adding finger-of-God effects causes $\gtrsim$10% deviations at $k \gtrsim 0.1\ h$ cMpc^{-1}, and that satellite contributions enter primarily at $k > 10\ h^{-1}\,$Mpc.

5.3 The revolution of 2023–2026

Since Pinetti et al.'s 2020 paper, the HI intensity mapping field has undergone a transformation from upper limits to high-significance detections.

MeerKAT. The landmark result is the **7.7σ cross-correlation** with WiggleZ galaxies at $z \approx 0.40$–0.46 (Cunnington et al., 2023), measuring $\Omega_{\rm HI} b_{\rm HI} r = (0.86 \pm 0.16) \times 10^{-3}$. Paul et al. (2023) achieved the first interferometric auto-power spectrum detection at $z = 0.32$ and 0.44, at 8.0σ and 11.5σ respectively on nonlinear scales ($0.3 < k < 8$ Mpc^{-1}). Carucci et al. (2025) confirmed the cross-correlation with improved multi-scale PCA cleaning, finding $\Omega_{\rm HI} b_{\rm HI} r = (0.93 \pm 0.17) \times 10^{-3}$ at $\sim 6\sigma$. The MeerKLASS deep field (MeerKLASS Collaboration, 2025) reached thermal noise of ~ 1.21 mK — the first time thermal noise is *subdominant to HI fluctuations* at $k \lesssim 0.15\,h^{-1}$ Mpc.

CHIME. CHIME detected 21-cm emission via stacking on eBOSS catalogues at $z = 0.78$–1.43 (CHIME Collaboration, 2023), achieving 7.1σ (LRG), 5.7σ (ELG), and **11.1σ (QSO)** detections. Cross-correlation with the Lyα forest at $z \approx 2.3$ yielded a **9σ** detection — the highest-redshift 21-cm emission measurement to date (CHIME Collaboration, 2024). The first **auto-power spectrum detection at** $z \sim 1$ with 12.5σ significance was announced in late 2025 (CHIME Collaboration, 2025), covering $k = 0.4$–$1.5\,h^{-1}$ Mpc at $z = 1.01$–1.34.

FAST. FAST remains in the systematics mitigation phase for cosmological intensity mapping, characterising $1/f$ noise (knee frequency $\sim 6 \times 10^{-4}$ Hz after SVD subtraction; Hu et al. 2021).

Does the Padmanabhan model hold up? The measured values of $\Omega_{\rm HI} \times b_{\rm HI}$ from MeerKAT and CHIME are *broadly consistent* with Padmanabhan et al. predictions on large scales. However, the CHIME $z \sim 1$ interpretation paper found 3.1–4.0σ tension with IllustrisTNG (CHIME Collaboration, 2026), suggesting that simulations underpredict nonlinear HI clustering in redshift space. The parametric halo model remains a viable framework, but extensions incorporating improved redshift-space distortion modelling and satellite contributions are needed for the precision era.

6 MeerKAT as an intensity mapping instrument

6.1 Array configuration

MeerKAT comprises **64 dishes** of **13.5 m diameter** (offset Gregorian, unblocked aperture) in the Upper Karoo, South Africa. About 70% of dishes lie within a dense 1-km core (Gaussian dispersion $\sim$300 m), with the remainder extending to maximum baselines of $\sim$8 km. Frequency coverage spans the UHF band (580–1015 MHz, HI at $z = 0.4$–1.45) and L-band (900–1670 MHz, HI at $z = 0.0$–0.58), each with 4096 frequency channels. In practice, the L-band cosmological window is severely limited by RFI to only 971–1023 MHz ($z \approx 0.39$–0.46), motivating the shift to UHF for MeerKLASS, which provides $\sim 40\times$ larger comoving volume (Cunnington et al., 2025).

6.2 Single-dish vs. interferometric mode

These modes access complementary angular scales.

Single-dish (auto-correlation) mode. Each dish operates as a scanning total-power radiometer, measuring large angular scales ($\ell \lesssim$ few hundred) critical for cosmological scales. The scanning strategy has dishes slewing at 5 arcmin/s in azimuth at constant elevation, with noise-diode injection every 20 s for relative gain calibration. The combined 64-dish system effectively multiplies integration time by 64.

Interferometric mode. 2016 baselines from 64 dishes resolve structure on small scales but filter out modes below $\ell_{\rm min}$ set by the shortest baseline ($\sim$29 m).

6.3 System temperature and noise

The noise model follows Pinetti et al. Eq. 3.18 for single-dish mode. The system temperature is

$$T_{\rm sys} = T_{\rm inst} + T_{\rm sky}, \qquad T_{\rm sky} = 60 \left(\frac{300\ {\rm MHz}}{\nu}\right)^{2.55} {\rm K}, \tag{15}$$

where $T_{\rm inst} \sim 18$–30 K (cryogenic receivers; $\sim$18 K at L-band centre per SARAO specifications, $\sim$30 K used in forecasts including spillover) and $T_{\rm sky}$ is dominated by Galactic synchrotron emission. At 1 GHz, $T_{\rm sky} \approx 3.3$ K gives $T_{\rm sys} \sim 20$–25 K; at 600 MHz, $T_{\rm sky} \approx 10$ K gives $T_{\rm sys} \sim 25$–40 K.

The thermal noise power spectrum scales as

$$P_N \propto \frac{T_{\rm sys}^2\, \Omega_{\rm survey}}{N_{\rm dish}\, t_{\rm obs}\, \delta\nu\, N_{\rm pol}}. \tag{16}$$

Achieved noise levels are 1.2–1.4× the theoretical limit: Wang et al. (2021) reported ~ 2 mK in the pilot survey, while the MeerKLASS deep field (2025) reached ~ 1.21 mK.

6.4 Beam and angular resolution

The single-dish beam is approximately Gaussian with FWHM (Pinetti et al. Eq. 3.17):

$$\theta_{\rm FWHM} \approx \frac{\lambda}{D_{\rm dish}}. \tag{17}$$

At 1 GHz, $\theta_{\rm FWHM} \approx 1.27°$; at 700 MHz, $\approx 1.82°$; at 580 MHz, $\approx 2.19°$. Frequency-dependent beam structure (chromaticity) couples with foregrounds and can bias signal reconstruction (Spinelli et al., 2021); the `eidos` package provides holographic beam models for more careful treatment.

6.5 Interferometric noise

The interferometric noise (Pinetti et al. Eq. 3.19) depends on the baseline density $n(u)$. At 1 GHz:

$$\ell_{\rm min} \approx 2\pi \times \frac{29 \text{ m}}{0.3 \text{ m}} \approx 607, \qquad \ell_{\rm max} \approx 2\pi \times \frac{8000}{0.3} \approx 167{,}000. \tag{18}$$

The dense core provides excellent short-baseline sensitivity, while the sparse outer baselines probe very small angular scales. The crossover multipole where interferometric noise beats single-dish noise occurs at $\ell \sim 200$–600, depending on frequency. For MeerKAT, cosmological scales ($k \lesssim 0.2\ h^{-1}\,$Mpc) are accessible only in single-dish mode; the interferometer covers the nonlinear regime.

6.6 MeerKLASS status update (2026)

The MeerKAT Large Area Synoptic Survey (Santos et al., 2017) originally proposed ~ 4000 deg^2 over ~ 4000 hours. As of early 2026, MeerKLASS has accumulated > 650 hours of UHF-band data reaching nearly 3000 deg^2, making it the largest spectroscopic survey of large-scale structure in the Southern hemisphere. Under the Extended Large Project (XLP) allocation, the survey targets 10,000 deg^2 by 2028 with ~ 2500 total hours. The **MeerKAT+ upgrade** adds 13–16 SKA-design dishes (first antenna handed over February 2024), extending the maximum baseline to ~ 17 km and increasing raw sensitivity by $\sim 50\%$. MeerKAT+ will eventually integrate into **SKA1-MID** (197 dishes total).

7 Foregrounds: enormous but spectrally distinguishable

7.1 The scale of the problem

Astrophysical foregrounds at MeerKAT frequencies are **4–5 orders of magnitude brighter** than the HI cosmological signal. Galactic synchrotron emission dominates, with brightness temperatures of ~ 1–6 K at high Galactic latitudes at 1 GHz (tens to hundreds of K near the plane), following a spectral index $\beta \approx -2.5$ to -2.9. Galactic free-free emission contributes ~ 1–10 K with a flatter spectral index $\beta \approx -2.1$. Extragalactic point sources (radio galaxies, AGN) add both clustered and Poisson components. The total foreground brightness is of order **Kelvin**, versus an HI signal of order **0.1–0.3 mK rms**.

7.2 Why removal is possible

All dominant foregrounds have *smooth power-law spectra* that vary slowly with frequency, while the cosmological 21-cm signal has rich structure along the frequency (line-of-sight) direction. Foregrounds therefore occupy only a few modes in the frequency dimension, making them separable in principle. In interferometric data, the chromatic response of baselines confines foreground power to a **wedge-shaped region** in $(k_\perp, k_\parallel)$ Fourier space, leaving a "clean window" at high $k_\parallel$ for signal extraction.

The primary limitations are: (i) **frequency-dependent beam** (chromaticity) that mixes foreground power into signal modes; (ii) **calibration errors** that introduce spectral structure mimicking HI; and (iii) **polarisation leakage** combined with Galactic Faraday rotation that creates non-smooth spectral features.

7.3 Removal techniques

The MeerKAT community employs several approaches.

PCA/SVD (blind). Decomposes the data matrix (frequency $\times$ pixel) into eigenmodes, removing the first $N_{\rm fg} = 4$–30 modes identified as foregrounds. Cunnington et al. (2023) used $N_{\rm fg} = 30$. **Multi-scale PCA** (Carucci et al., 2025) applies PCA with different $N_{\rm fg}$ at different angular scales, optimising foreground removal while preserving cosmological signal.

Gaussian Process Regression (GPR). Models data as smooth foreground (long frequency correlation length) plus signal+noise (short correlation length). Better at recovering radial ($k_\parallel$) power spectrum than PCA (Soares et al., 2022).

SDecGMCA. Sparse component separation that jointly performs beam deconvolution and source separation (Gkogkou et al., 2025), achieving $\lesssim 5\%$ accuracy at $\ell = 10$–200 under realistic beam conditions — outperforming standard methods.

Foreground avoidance. Used in interferometric mode (delay spectrum technique; Paul et al. 2023), measuring the power spectrum only in the foreground-free region of Fourier space.

All methods cause some **signal loss**, quantified via the transfer function

$$T(k) = \frac{P_{\rm recovered}(k)}{P_{\rm injected}(k)}, \tag{19}$$

constructed by injecting mock HI signal into real data, re-running the cleaning pipeline, and measuring the recovery fraction. Despite $\sim 50\%$ signal loss on the largest scales, the power spectrum can be reconstructed to sub-percent accuracy using this transfer function approach.

7.4 The cross-correlation advantage for HI $\times$ gamma rays

This is perhaps the most important point for the Pinetti et al. analysis. Foregrounds in HI maps (Galactic synchrotron, free-free) and foregrounds in Fermi-LAT gamma-ray maps (Galactic diffuse gamma-ray emission from cosmic-ray interactions) have **completely different physical origins** and are therefore largely **uncorrelated**. Foreground residuals in one map do not systematically bias the cross-correlation measurement — they increase variance but not the mean of the cross-power estimate.

This has been demonstrated observationally: Cunnington et al. (2019) showed that foregrounds have little impact on optical spectroscopic cross-correlations, with reconstructed cross-power within 8.5% even at scales below beam resolution. The practical success of cross-correlation detections — MeerKAT $\times$ WiggleZ at 7.7σ, CHIME $\times$ eBOSS at 11.1σ, CHIME $\times$ Lyα at 9σ despite foreground residuals 6–10$\times$ thermal noise — constitutes strong empirical evidence.

7.5 Spurious cross-correlation risks

The primary concern is **radio-loud AGN that are also gamma-ray emitters**. The Fermi-LAT 4FGL-DR4 catalogue contains 7194 sources, of which $> 60\%$ are blazars — and essentially all blazars are radio-loud by definition. Mitigation requires: (i) masking all 4FGL source positions and their error ellipses from both maps; (ii) masking known bright radio sources from continuum catalogues (NVSS, SUMSS, VLASS) above a flux threshold; and (iii) statistically estimating the residual contribution from unresolved sources below detection thresholds using source-count models.

8 From raw MeerKAT data to science-ready intensity maps

8.1 Calibration

Standard interferometric calibration achieves bandpass stability within $\sim 3\%$ over 3 hours. Intensity mapping requires stability at the $\sim 10^{-5}$ level to separate the μK HI signal from Kelvin-scale foregrounds. The MeerKLASS single-dish pipeline (Wang et al., 2021) models total-power data as a sum of components: calibrator temperature, diffuse foreground, elevation-dependent terrestrial emission, noise-diode signal, and receiver residual. Calibration uses periodic noise-diode injection (1.8 s every 20 s) for relative gain tracking, with absolute calibration from celestial sources (3C 273, Pictor A) every ~ 1.5 hours.

$1/f$ **noise.** Correlated temporal gain fluctuations are the critical systematic for single-dish mode. Li et al. (2021) measured a knee frequency of $\sim 3 \times 10^{-3}$ Hz after 2-SVD-mode subtraction, implying adequate gain stability over timescales of a few hundred seconds. The $1/f$ noise is highly correlated in frequency and hence removable via SVD/PCA, but aggressive cleaning risks signal loss.

Polarisation leakage. Stokes $Q, V \rightarrow I$ leakage combined with Faraday rotation introduces non-smooth spectral structure requiring careful correction.

8.2 RFI excision

Despite the Karoo's radio-quiet zone designation, RFI from telecommunications satellites (especially in L-band ~ 1150–1300 MHz), distant mobile towers, and aircraft transponders remains significant. The Wang et al. (2021) pipeline uses a 3-round flagging approach: (i) strong RFI via the `SEEK` package (SumThreshold algorithm), flagging 40–60% across the full L-band but only **5–25% in target cosmological windows**; (ii) per-channel outlier removal in time-ordered data; and (iii) map-based identification of residual weak RFI. The UHF band suffers less severe contamination, with RFI restricted to narrow frequency chunks. Additional tools include Tricolour (GPU-accelerated flagger) and deep learning approaches (R-Net architecture).

8.3 Map-making

Calibrated time-ordered data (Stokes I in Kelvin) are projected into sky maps using optimal map-making:

$$\hat{m} = \left(\mathbf{A}^\top \mathbf{N}^{-1} \mathbf{A}\right)^{-1} \mathbf{A}^\top \mathbf{N}^{-1} \mathbf{d}, \tag{20}$$

where $\mathbf{A}$ is the pointing matrix and $\mathbf{N}$ the noise covariance. The MeerKLASS pilot used $0.3°$ square pixels in flat-sky approximation; recent work employs Cartesian regridding of spherical maps to avoid wide-sky biases. HEALPix pixelisation is used in simulations and some analysis frameworks. Foreground subtraction (PCA/GPR/GMCA) then operates on these frequency–pixel data cubes. The transfer function is computed at this stage via mock signal injection.

8.4 Software ecosystem

The MeerKAT community has developed a rich set of publicly available tools. **CARACal** (formerly MeerKATHI) is an end-to-end containerised calibration and imaging pipeline for interferometric data, using the Stimela framework (Józsa et al., 2020). **katdal** is a Python data access library for MeerKAT Visibility Format (MVF) files, handling HDF5/RDB chunk stores and conversion to CASA MeasurementSets. **katcali** is the MeerKLASS-specific single-dish calibration pipeline for auto-correlation intensity mapping, handling RFI flagging, noise-diode calibration, absolute calibration, and map-making. **oxkat** provides semi-automated calibration and imaging scripts covering the full workflow from flagging through self-calibration (Heywood, 2020). **katbeam** and **eidos** provide primary beam models (simplified cosine-taper and holographic, respectively). **MeerFish** is a Fisher forecast code released alongside Cunnington et al. (2025).

8.5 Public data products

MeerKLASS has released the **first public HI intensity mapping data cube** from the 2019 pilot survey via `https://meerklass.org/data.html`, covering $153° < \mathrm{RA} < 172°$, $-1° < \mathrm{Dec} < 8°$ as calibrated intensity cubes after map-making but before foreground removal. UHF and L-band continuum data releases provide complementary radio source catalogues. MIGHTEE deep-field data in COSMOS, E-CDFS, ELAIS-S1, and XMM-LSS fields support interferometric intensity mapping analyses.

9 Conclusion: a framework that has aged well but needs updating

The Pinetti et al. (2020) framework rests on a solid theoretical foundation that has been broadly validated by the subsequent explosion of observational results. Three key findings emerge from this assessment.

First, the Padmanabhan et al. (2017) HI halo model remains the most practical analytic framework for cross-correlation predictions, but it requires extensions for the precision era — particularly improved modelling of nonlinear redshift-space distortions, satellite contributions, and the 3.1–4.0σ tension with IllustrisTNG identified by CHIME at $z \sim 1$.

Second, the cross-correlation advantage for foreground mitigation has been *spectacularly confirmed* by multiple experiments: MeerKAT, CHIME, and GBT cross-correlations all succeed even when auto-correlations remain foreground-dominated. This bodes well for the HI $\times$ gamma-ray cross-correlation, where the physical disconnect between radio and gamma-ray foregrounds is even stronger than between radio and optical foregrounds.

Third, the dominant uncertainty in the predicted signal amplitude comes from the product $\Omega_{\rm HI} \times b_{\rm HI}$, constrained to ~ 15–20% at $z \sim 0.4$ by MeerKAT but still poorly known at $z > 1$. For this analysis, the specific parameter values adopted by Pinetti et al. (particularly $v_{\rm c,0} = 101.61$ km s^{-1}) should be carefully checked against the original Padmanabhan et al. tables and tested for consistency with the latest MeerKAT and CHIME constraints.

The MeerKLASS UHF-band survey, now exceeding 3000 deg^2, combined with 14 years of Fermi-LAT data (4FGL-DR4), places the first actual measurement of the HI $\times$ gamma-ray cross-correlation within realistic reach.

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